

CSE570:MACHINE LEARNING WITH PYTHON

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

- CO1 :: understand Python-based tools and libraries used for machine learning.
- CO2 :: apply data preprocessing, cleaning, encoding, and scaling techniques for ML datasets.
- CO3 :: analyze datasets using Pandas, NumPy, visualization, and exploratory data analysis.
- CO4 :: understand supervised ML models for regression and classification problems.
- CO5 :: apply unsupervised learning techniques such as clustering and dimensionality reduction.
- CO6 :: apply Python libraries to build, tune, save, and interpret machine learning models for predictive analytics tasks.

Unit I

Python fundamentals and Data Preparation for Machine Learning : Python essentials, NumPy arrays, Pandas DataFrames, data loading, missing value handling, encoding categorical variables, feature scaling, train-test split

Unit II

Exploratory Data Analysis and Feature Engineering : EDA, visualization, correlation analysis, outlier detection, feature selection, feature transformation, data imbalance handling, data preprocessing pipelines

Unit III

Supervised Machine Learning – Regression : Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, Regularization, Ridge, Lasso, Elastic Net, regression evaluation metrics such as MAE, MSE, RMSE, R squared

Unit IV

Supervised Machine Learning – Classification : Logistic Regression, KNN, Decision Tree, Random Forest, SVM, Naïve Bayes, classification metrics, confusion matrix, accuracy, precision, recall, F1-score, ROC-AUC

Unit V

Unsupervised Learning and Dimensionality Reduction : K-Means Clustering, Hierarchical Clustering, DBSCAN, PCA, clustering evaluation, silhouette score, real-world clustering applications

Unit VI

Model Evaluation, Optimization and Recent Trends : Cross-validation, hyperparameter tuning, GridSearchCV, RandomizedSearchCV, model persistence, ML pipelines, introduction to ensemble learning, XGBoost/LightGBM, basics of explainable AI, ethical issues in ML

List of Practicals / Experiments:

List Of Practicals

- Implement data preparation and preprocessing techniques using Python, including data loading, handling missing values, encoding categorical variables, feature scaling, and train-test data splitting.
- Perform Exploratory Data Analysis (EDA), Correlation Analysis, Outlier Detection, Feature Selection, Feature Transformation, and Handle Imbalanced Data.
- Implement and Compare Simple Linear Regression, Multiple Linear Regression, and Polynomial Regression using appropriate evaluation metrics (MAE, MSE, RMSE, R^2).
- Implement Ridge Regression, Lasso Regression, and Elastic Net Regression and compare their performance.
- Implement and Compare Classification Algorithms including Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree.
- Implement Random Forest, Support Vector Machine (SVM), and Naïve Bayes classifiers and evaluate their performance using Confusion Matrix, Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Perform Clustering using K-Means, Hierarchical Clustering, and DBSCAN, and evaluate the clustering performance using Silhouette Score.

- Apply Principal Component Analysis (PCA) for Dimensionality Reduction and analyze its impact on model performance.
- Perform Model Evaluation and Optimization using Cross-Validation, GridSearchCV, and RandomizedSearchCV, and save the best-performing model.

Text Books:

1. PYTHON FOR DATA SCIENCE by MOHD. ABDUL HAMEED, WILEY

References:

1. FOUNDATIONAL PYTHON FOR DATA SCIENCE by KENNEDY BEHRMAN, PEARSON
2. DATA SCIENCE AND MACHINE LEARNING USING PYTHON by REEMA THAREJA, MC GRAW HILL